SP19_340_Final (6)

Math 340-004: Elementary Matrix and Linear Algebra

MIDTERM TWO (Note: This is a typo, and should be **Final Exam**)

May 5, 2019

TABLE OF DISCUSSION SECTIONS

Section	Meets		Leader
361	M	8:50 - 9:40 am	Ongay Valverde
362	M	9:55 - 10:45 am	Ongay Valverde
363	M	11:00-11:50 am	Mohamed
364	M	12:05-12:55 pm	Mohamed
365	M	1:20-2:10 pm	Mohamed

Name: Key ______ Section#: ______ NetID: ______

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______

READ THE FOLLOWING INFORMATION.

- This is a **90-minute** exam. It consists of 6 problems on 7 pages. It is your responsibility to make sure that you have a complete exam.
- ~~You may use a standard scientific calculator, but not one which has sizeable stored memory.~~
- Books, notes, and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. ***Do not unstaple or remove pages as they can be lost in the grading process.***
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.

1. (16 points) For each of the following, please give an appropriate definition for the word or phrase in *italics*.

   (a) The set W is a *subspace* of a vector space V.

   W is a nonempty subset of V which is itself a vector space using the same addition/scalar mult. as V.

   (b) The set S is a *basis* for a vector space V.

   S spans V, and S is linearly independent.

   (c) The function L is a *linear transformation* from V to W.

   $L(\vec{u}+\vec{v}) = L(\vec{u}) + L(\vec{v})$ for all $\vec{u}, \vec{v}$ in V.

   $L(c\vec{u}) = cL(\vec{u})$ for all $\vec{u}$ in V, c in $\mathbb{R}$.

   (d) The linear transformation has *rank* n.

   Dim range(L) = n <u>or</u> a matrix representing L has rank n.

   (e) ~~The *geometric multiplicity* of an eigenvalue of the matrix A.~~

   # of elements in a basis for the eigenspace.

   (f) The set C is the *null space* of a matrix A.

   set of vectors $\vec{v}$ in $\mathbb{R}^n$ where $A\vec{v} = \vec{0}$

   (g) A is a *singular* matrix.

   A is <u>not</u> invertible, i.e. there is no matrix B where $AB = I_n$.

   (h) S is an *orthonormal* basis for the inner product space V.

   Each pair of elements in S is orthogonal and every vector in S has length 1.

2. (12 points) Let

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 7 & -5 \\ 3 & 9 & x \end{bmatrix}$$

where x is some real number. Compute the following (if possible). If the computation is not possible, explain why.

(a) Row reduce A to an upper triangular matrix.

$$A \xrightarrow[-3r_1+r_3 \to r_3]{-2r_1+r_2 \to r_2} \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 6+x \end{bmatrix} \quad \text{Done !!}$$

$$\underset{B}{\shortparallel}$$

(b) What is the determinant of A?

can use B. $\det(A) = \det(B)$ b/c we used $r_i + c r_j \to r_i$ row op.

$\det(A) = \det(B) = 6 + x$

(c) Find all values of x which make the matrix singular.

want: $\det(A) = 0$

$6 + x = 0$

$\boxed{x = -6}$

(d) Find all values of x for which the rank of the matrix is 3.

rank 3 $\Rightarrow$ since A is 3×3, this means A is invertible.

so $\det(A) \neq 0$

want $\boxed{x \neq -6}$

3. (9 points) Let $L : P_3 \to P_3$ by

$$L(p) = p'(t) + p''(t).$$

(a) Let $\gamma = (t^3, t^2, t, 1)$ be the standard basis for P_3. Give a matrix representation for L under this basis.

$L(t^3) = 3t^2 + 6t \to \begin{bmatrix} 0 \\ 3 \\ 6 \\ 0 \end{bmatrix}$

$L(t^2) = 2t + 2 \to \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix}$

$L(t) = 1 + 0 \to \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$L(1) = 0 + 0 \to \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

matrix rep: $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 6 & 2 & 0 & 0 \\ 0 & 2 & 1 & 0 \end{bmatrix}$

(b) Find a basis for ker L. Use matrix rep:

$$A \xrightarrow[r_2 \leftrightarrow r_1]{\frac{1}{3}r_2 \to r_2,} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & 2 & 0 & 0 \\ 0 & 2 & 1 & 0 \end{bmatrix} \xrightarrow{-6r_1 + r_3 \to r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 2 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{\frac{1}{2}r_3 \to r_3 \\ r_2 \leftrightarrow r_4, \\ r_2 \leftrightarrow r_3}]{-r_3 + r_4 \to r_4,} \begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & x_4 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

pivot cols.

$x_4 = r$
$x_3 = 0$
$x_2 = 0$
$x_1 = 0$

$r \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \to r(1)$

Basis for ker L: $\{1\}$

(c) Find a basis for range L.

Pick up the pivots!

$\begin{bmatrix} 0 \\ 3 \\ 6 \\ 0 \end{bmatrix} \to 3t^2 + 6t$, $\begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix} \to 2t + 2$, $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \to 1$

Basis: $\{3t^2 + 6t, 2t + 2, 1\}$

4. (12 points) Consider the inner product space $P_2[0,1]$ of polynomials of degree 2 on the interval $[0,1]$ with the inner product

$$(p,q) := \int_0^1 pq\,\mathrm{d}x$$

(a) Find a unit vector parallel to $p(t) = t$.

$$\|t\| = \sqrt{\int_0^1 t\cdot t\,dt} = \sqrt{\int_0^1 t^2\,dt} = \sqrt{\frac{t^3}{3}\Big|_0^1} = \sqrt{\frac{1}{3}}$$

answer: $\boxed{\frac{t}{\sqrt{1/3}} = \sqrt{3}\,t}$

~~(b) Let $S = \text{span}\{1, t+2\}$. Find an orthonormal basis for S.~~

see section 5.4. (we don't cover this)

(c) With S as in part (b), ~~compute $S^\perp$.~~ Find a basis for the set of vectors orthogonal to S.

want: q w/ $(1,q)=0$ and $(t+2,q)=0$. $q = at^2+bt+c$

$$(1,q) = \int_0^1 at^2+bt+c\,dt = \frac{at^3}{3}+\frac{bt^2}{2}+ct\Big|_0^1 = \boxed{\frac{a}{3}+\frac{b}{2}+c=0} \text{ Eq.1}$$

$$(t+2,q) \to \int_0^1 (t+2)(at^2+bt+c)\,dt = \int_0^1 at^3+bt^2+ct+2at^2+2bt+2c\,dt = \frac{at^4}{4}+\frac{(b+2a)t^3}{3}+\frac{(2b+c)t^2}{2}+2ct\Big|_0^1$$

$$= \frac{a}{4}+\frac{b+2a}{3}+\frac{2b+c}{2}+2c=0$$

$$\boxed{\frac{11a}{12}+\frac{4b}{3}+\frac{5}{2}c=0} \text{ Eq.2}$$

~~(d) With S as in part (b), find the vector v in S so that $\|v - x^2\|$ is as small as possible.~~

(Part c):

$$\left[\begin{array}{ccc|c} 1/3 & 1/2 & 1 & 0 \\ \frac{11}{12} & 4/3 & \frac{5}{2} & 0 \end{array}\right] \xrightarrow{-\frac{11}{4}r_1+r_2\to r_2} \left[\begin{array}{ccc|c} 1/3 & 1/2 & 1 & 0 \\ 0 & -1/24 & -1/4 & 0 \end{array}\right]$$

$-24r_2 \to r_2$, $3r_1 \to r_1$

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$
$$\left[\begin{array}{ccc|c} 1 & 3/2 & 3 & 0 \\ 0 & 1 & 6 & 0 \end{array}\right]$$

$c = r$
$b = -6r$

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$a = -3r - \frac{3}{2}b$
$= -3r - \frac{3}{2}(-6r) = 6r$

q is of the form $\underbrace{6rt^2 - 6r + r = r(6t^2 - 6r + 1)}_{S^\perp}$

5. (6 points) Let

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(a) What is the characteristic polynomial of A?

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda-3 & 2 & 0 \\ 2 & \lambda-3 & 0 \\ 0 & 0 & \lambda-5 \end{bmatrix} = (\lambda-5)((\lambda-3)(\lambda-3)-4)$$

$$= (\lambda-5)(\lambda^2 - 6\lambda + \overbrace{9-4}^{5})$$

$$= (\lambda-5)(\lambda^2-6\lambda+5)$$

$$= (\lambda-5)(\lambda-5)(\lambda-1)$$

(b) Is A diagonalizable? Why or why not?

$\lambda = 5, 5, 1 \rightarrow$ not distinct, so maybe?

Check $\lambda=5$:

$$\begin{bmatrix} 5-3 & 2 & 0 \\ 2 & 5-3 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{then } \frac{1}{2}r_1 \to r_1]{-r_1+r_2\to r_2} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

let $x_3 = r$

$x_2 = s$

$x_1 = -s$

$$\begin{bmatrix} -s \\ s \\ r \end{bmatrix}$$

Eigenspace of $\lambda=5$:

$$\begin{bmatrix} -s \\ s \\ r \end{bmatrix} = s\underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}} + r\underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$$

2 linearly indep. eigenvectors

$\lambda = 1$ will also give us another linearly indep. eigenvector $\Rightarrow$ we have 3 linearly indep. eigenvectors $\Rightarrow$ A is diagonalizable.

6. (16 points) Indicate (clearly) if the following statement is true or false. In either case give a short explanation why.

(a) A linear system of three equations can have exactly two solutions.

False. A linear system can have ① a unique solution ② No solution or ③ Infinitely many solutions.

(b) A product of invertible matrices is non-singular.

True, b/c $\det(AB) = \det(A)\det(B)$ → If each of these is nonzero, $\det(\text{product}) \neq 0$, which means invertible.

(c) $\det(AB^TA^{-1}) = \det(B)$.

True, $\det(AB^TA^{-1}) = \cancel{\det(A)}\underbrace{\det(B^T)}_{=\det(B)}\underbrace{\det(A^{-1})}_{=\frac{1}{\cancel{\det(A)}}} = \det(B)$.

(d) A linearly independent set can have a linearly dependent subset.

False, Any subset of a linearly independent set is linearly indep. (think pivots)

(e) If v is orthogonal to both u and w then v is orthogonal to every vector in span$\{u, w\}$.

True. Let $\vec{x}$ be in the span of $\vec{u}, \vec{w} \Rightarrow \vec{x} = a\vec{u} + b\vec{w}$. If $(\vec{v}, \vec{u}) = 0$ and $(\vec{v}, \vec{w}) = 0$,

$(\vec{v}, \vec{x}) = (\vec{v}, a\vec{u} + b\vec{w}) = (\vec{v}, a\vec{u}) + (\vec{v}, b\vec{w}) = a(\vec{v}, \vec{u}) + b(\vec{v}, \vec{w})$
$= (a)(0) + (b)(0) = 0.$

(f) There is a linear transform $T : \mathbb{R}^3 \to \mathbb{R}^2$ which has a trivial kernal.

False. $\underbrace{\dim \ker T}_{\text{Needs to be at least } 1.} + \underbrace{\dim \operatorname{range} T}_{\text{max is } \dim(\mathbb{R}^2) = 2.} = \dim(\mathbb{R}^3) = 3.$

(g) There is a linear transform $T : \mathbb{R}^2 \to \mathbb{R}^3$ which has trivial kernal.

True. For example, $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$

(h) Every 2×2 matrix is invertible.

False. ex: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \to \det\left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\right) = 1 \cdot 1 - 1 \cdot 1 = 0.$